

Chapter | 10 | Formatting Reports and Proposals



>>> Chapter Objectives

In this chapter, students will

- Learn how to format informal and formal documents like reports and proposals
- Learn when to use informal and formal formats for longer documents
- Read about sample situations in which informal and formal reports and proposals are written
- Learn guidelines to format informal documents to communicate a professional image and to make information clear for readers
- Learn the nine parts of formal documents
- Learn guidelines to format formal documents to communicate a professional image and to create navigation elements and organize information to help readers find the information that they need
- Read and analyze model reports and proposals

Kurt Fleisch, M-Global's director of marketing, was given an interesting assignment two months ago. Jim McDuff asked him to take a long, hard look at the company's clients. Are they satisfied with the service they receive? Do they routinely reward M-Global with additional work? Are there any features of the company, its employees, or its services that frustrate them? What do they want to see changed? In other words, Kurt was asked to step back from daily events and evaluate the company's level of service. He attacked the project in five stages:

1. He searched Internet sources to identify a target rate for client retention among companies like M-Global.
2. He designed and sent out a survey to all recent and current clients.
3. He followed up on some of the returned surveys with phone and personal interviews.
4. He evaluated the data he collected.
5. He decided to use a formal format for his report on the results of his study. In addition to going to all corporate and branch managers, Fleisch's report later served as a basis for a proposal for in-house training sessions called Quality at M-Global.

Like Kurt Fleisch, you will write a number of reports and proposals during your career. Most will be written collaboratively with colleagues; others will be your responsibility. All will require major efforts at planning, organizing, drafting, and revising. Reports are used to record activities and share research for decision making, and proposals, like reports, are part of the decision-making process. Reports and proposals are more complex than the definitions, descriptions, instructions, and process explanations that are discussed in Chapter 7 and Chapter 8. In fact, those genres may be included as sections of reports and proposals. You may write reports and proposals for readers within your organization, or you may write these documents for outside readers such as people in other

organizations, government agencies, or even the general public.

Reports and proposals should be adapted to their audiences and purposes, but they are formatted in two basic ways—as informal or as formal documents. (Chapter 11 and Chapter 12 offer strategies for tailoring reports and proposals to specific purposes.) Informal reports and proposals can be formatted as letters (for outside readers) or as memos (for inside readers). Formal reports and proposals include the same basic elements whether they are written for internal or external audiences.

Often, you will not have to choose whether to use an informal or a formal format. The format will either be part of the project assignment, or it will be obvious from the length and complexity of the document. Sometimes, however, you will need to choose the format. In the M-Global example at the beginning of the chapter, Kurt decided to use a formal format for his report. He did so because his document was fairly complex, with a large number of tables, charts, and graphs. His report will also be read by people with differing interests and areas of expertise. He used a formal format for his proposal, as well, since he wanted to recommend a new training program that would require a significant investment; he decided that the formal format would signal a level of importance to his recommendation.

This chapter provides guidelines to help you format reports and proposals so that they meet readers' expectations and so that readers can find the information that they need. We begin with basic definitions of the two basic formats. This text uses the following working definition of informal documents:

Informal document: A somewhat short document, usually no longer than five pages of text, not including attachments. It has more substance than a simple letter or memo but is presented in letter or memo format. It can be directed to readers either outside or inside your organization. If outside, it may be called a *letter report* or *letter proposal*; if inside, it may be called a *memo report* or *memo proposal*.

Although informal reports are the most common report in business writing, formal reports become a larger part of your writing as you move along in your career. Many proposals also use a formal format. This text uses the following working definition:

Formal document: A formal document covers complex projects and is directed to readers at different technical levels. Although not defined by length, a formal document usually contains at least six pages of text, not including appendixes. It can be directed to readers either inside or outside your organization. Often bound, it usually includes the following separate parts: (1) cover/title page, (2) letter/memo of transmittal, (3) table of contents, (4) list of illustrations, (5) executive summary, (6) introduction, (7) discussion sections, and (8) conclusions and recommendations. (9) End material such as appendixes and bibliographies.

As noted previously, informal documents and formal documents look quite different. Early in your career, however, you may have trouble deciding which format to use. To help you decide, the two lists that follow briefly describe the characteristics of each format.

Informal reports and proposals have the following characteristics:

- Informal documents have a narrower focus, on a specific problem, situation, or event.
- Informal documents may be written by a team, but they are often written by a single author.
- Informal documents usually have few readers, or even just one reader.
- Informal documents are usually two to five pages long.
- Informal documents use letter (for external audiences) or memo (for internal audiences) format.
- Informal documents may be created in a preset form or a template.
- Informal documents use headings to help readers find information.
- Informal documents may include appendixes.

Formal reports and proposals have the following characteristics:

- Formal documents usually address complex problems, situations, or events.
- Formal documents are often written by a team.
- Formal documents usually are created for multiple readers at different technical levels.
- Formal documents generally include at least six pages of text.
- Formal documents are usually created for external audiences, although they may be used internally if the document is long and complex.
- Formal documents are often bound or presented in some kind of cover.
- Formal documents use headings, subheadings, and other navigational elements to help readers find information.
- Formal documents include front and back material, such as a title page, a table of contents, and appendixes.

The rest of this chapter includes information to help you decide when to use informal or formal formatting for your reports and proposals and provides guidelines for both basic formats.



>>> When to Use Informal Document Format

In your career, you will spend much of your time writing informal reports and proposals. Although they are short and easy to read like letters and memos, these informal documents have more substance, are longer, and thus require more organization signals than everyday correspondence. This section first shows you when to use informal documents in your career by describing some M-Global cases. Second, it provides 10 main writing guidelines that apply to both letter and memo documents.

Letter Reports and Proposals at M-Global

Written to people outside your organization, letter reports and proposals use the format of a business letter because of their brevity; however, they include more detail than a simple business letter. Following are some sample projects at M-Global that would require letter reports or proposals:

- **Seafloor study:** M-Global's Nairobi staff writes a preliminary report on the stability of the seafloor where an oil rig might be located off the coast of Africa. This preliminary study includes only a survey of information on file about the site. The final report, involving fieldwork, will be longer and more formal.
- **Environmental study:** M-Global's San Francisco staff reports to the local Sierra Club chapter on possible environmental effects of an entertainment park proposed for a rural area where eagles often nest. The project involves one site visit, interviews with a biologist, and some brief library research.
- **Proposal for training project:** M-Global's corporate training staff proposes changes in the training program of a large construction company. Courses that are described include technical writing, interpersonal communication, and quality management.
- **Sales proposal for asbestos removal:** Jane Wiltshire, asbestos department manager at M-Global's St. Paul office, regularly talks with owners of buildings that may contain asbestos. After an initial discussion with the head minister of First Street Church, she writes an informal sales proposal in which she offers M-Global's services in performing an asbestos survey of the church building. Specifically, she explains how M-Global will examine the structure for possible asbestos, gives a schedule for completing the survey and writing the final report, and proposes a lump-sum price for the project.

As these examples show, letter reports and proposals are the best format for projects with a limited scope. In addition, this informal format is a good sales strategy when dealing with customers greatly concerned about the cost of your work. When reading letter proposals, they realize—consciously or subconsciously—that these documents cost them less money than formal proposals. Your use of letter proposals for small jobs shows

sensitivity to their budget and may help gain their repeat work. See Model 10–1 on pages 330–331 for a letter report based on a small project at M-Global.

Memo Reports and Proposals at M-Global

Memo reports and proposals are the informal documents that go back and forth among M-Global's own employees. Although in memo format, they include more technical detail and are longer than routine memos. These situations at M-Global show the varied contexts of memo reports and proposals:

- **Need for testing equipment:** Juan Watson, a lab technician in the Denver office, evaluates a new piece of chemical testing equipment for his department manager, Wes Powell. Powell discusses the report with his manager.
- **Personnel problem:** Werner Hoffman, a field engineer in the Munich office, writes to his project manager, Hans Schulman, about disciplinary problems with a field hand. Hoffman discusses the report with his manager and with the personnel manager.
- **Report on training session:** Pamela Martin, a field engineer in St. Louis, reports to her project manager, Mel Baron, on a one-week course she took in Omaha on new techniques for removing asbestos from buildings. Baron circulates the report to his office manager, Ramsey Pitt, who then sends copies to the manager of every company office, because asbestos projects are becoming more common throughout the firm.
- **Proposal for structural design and analysis equipment:** Meg Stevens, a civil engineer at the Denver office, writes an in-house proposal to the construction manager, Elvin Lipkowsky, in which she proposes that the company purchase a new structural design and analysis system. Her proposal includes a description of equipment that she recently saw demonstrated at a conference of civil engineers.
- **Proposal for retaining legal counsel:** Jake Washington, an employment specialist in the Human Resources Department in the Baltimore office, writes an in-house proposal to Karrie Camp, Vice President for Human Resources. In it he proposes that the company retain legal counsel on a half-time basis (20 hours a week). In his position at M-Global, Jake uses outside legal advice in dealing with new hiring laws, unemployment compensation cases, affirmative action regulations, and occasional lawsuits by employees who have been fired. He is proposing that the firm retain regular half-time counsel, rather than dealing with different lawyers as is done now.

These five documents require enough detail to justify writing memo reports or proposals rather than simple memos. As for audience, each document goes directly to one reader, and it may be discussed with readers at high levels within the company. That means good memo reports and proposals can help advance your career. Model 10–2 on pages 332–333 provides an annotated example of a memo proposal from a small non-profit organization.

>>> General Guidelines for Informal Document Format

Following are 10 guidelines that focus mainly on informal document format.

>> Informal Document Guideline 1: Plan Well Before You Write

Like other chapters in this book, this section emphasizes the importance of the planning process. Complete the Planning Form at the end of the book for each assignment in this chapter, as well as for informal documents you write in your career. Before you begin writing a draft, use the Planning Form to record specific information about these points:

- The document's purpose
- The variety of readers who will receive the document
- The needs and expectations of readers, particularly decision makers
- An outline of the main points to be covered in the body
- Strategies for writing an effective document

>> Informal Document Guideline 2: Use Letter or Memo Format

Model 10–1 shows that letter reports and proposals follow about the same format as typical business letters (see Chapter 6). For example, both are produced on letterhead and both often include the reader's name, the date, and the page number on all pages after the first. Yet the format of letter reports and proposals differs from that of letters in the following respects:

- The greeting is sometimes left out or replaced by an attention line, especially when your letter report or proposal will go to many readers in an organization.
- A title often comes immediately after the inside address. It identifies the specific project covered in the document. You may have to use several lines because the project title should be described fully, in the same words that the reader would use.
- Spacing between lines might be single, one-and-one-half, or double, depending on the reader's preference.

Model 10–2 shows the typical format for a memo report or proposal. Like most memos, it includes Date/To/From/Subject information at the top and has the recipient's name, the date, and the page number on every page after the first. Also, memos and memo reports and proposals have a subject line that should engage interest, give readers their first quick look at your topic, and be both specific and concise—for example, “Fracture Problems with Molds 43-D and 42-G” is preferable to “Problems with Molds.” Because memo reports and proposals are usually longer than memos, they tend to contain more headings than routine memos.



Franz Pfluegl/Dreamstime.com

>> Informal Document Guideline 3: Make Text Visually Appealing

Your informal report or proposal must compete with other documents for each reader's attention. Following are three visual devices that help get attention, maintain interest, and highlight important information:

- Bulleted points for short lists (like this one)
- Numbered points for lists that are longer or that include a list of ordered steps
- Frequent use of headings and subheadings

Headings are particularly useful in memo and letter reports and proposals. As Models 10–1 and 10–2 (pp. 330–333) show, they give readers much-needed visual breaks. Because informal documents have no table of contents, headings also help readers locate information quickly. (Chapter 5 gives more detail on headings and other features of page design.)

>> Informal Document Guideline 4: Use the ABC Format for Organization

Headings and lists attract attention, but these alone do not keep readers interested. You must also organize information effectively. Most technical documents, including informal documents, follow what this book calls the *ABC format*. This approach to organization includes three parts: (1) *Abstract*, (2) *Body*, and (3) *Conclusion*.

Abstract, *body*, and *conclusion* are only generic terms. They indicate the types of information included at the beginning, middle, and end of your documents—not necessarily the exact headings you will use. The next four guidelines give details on the ABC format as applied to memo and letter reports and proposals.

ABC Format: Informal Documents

- **ABSTRACT:** Start with a capsule version of the information most needed by decision makers.
- **BODY:** Give details in the body of the document, where technical readers are most likely to linger a while to examine supporting evidence.
- **CONCLUSION:** Reserve the end of the document for a description or list of findings, conclusions, or recommendations.

>> Informal Document Guideline 5: Create the Abstract as an Introductory Summary

Abstracts should give readers a summary, the “big picture.” This text suggests that in informal documents, you label this overview *Introduction*, *Summary*, or *Introductory Summary*, terms that give the reader a good idea of what the section contains. (You also have the option of leaving off a heading label, in which case your first few paragraphs will contain the introductory summary information, followed by the first body heading of the document.)

In letter reports and proposals, the introductory summary comes immediately after the title; in memo reports and proposals, it comes after the subject line. Note that informal

documents do not require long, drawn-out beginnings; just one or two paragraphs in this first section give readers three essential pieces of information:

1. **Purpose** for the document—Why are you writing it?
2. **Scope** statement—What range of information does the document contain?
3. **Summary** of essentials—What main information does the reader most want or need to know?

>> Informal Document Guideline 6: Put Important Details in the Body

The body section provides details needed to expand on the outline presented in the introductory summary. If your document goes to a diverse audience, managers often read the quick overview in the introductory summary and then skip to the last section of the document. Technical readers, however, may look first to the body section(s), where they expect to find supporting details presented in a logical fashion. In other words, here is your chance to make your case and to explain points thoroughly.

Yet the discussion section is no place to ramble. Details must be organized so well and put forward so logically that the reader feels compelled to read on. Following are three main suggestions for organization:

- **Use headings generously.** Each time you change a major or minor point, consider whether a heading change would help the reader. Informal documents should include at least one heading per page.
- **Precede subheadings with a lead-in passage.** Here you mention the subsections to follow, before you launch into the first subheading. (For example, “This section covers these three phases of the field study: clearing the site, collecting samples, and classifying samples.”) This passage does for the entire section exactly what the introductory summary does for the entire document—it sets the scene for what is to come by providing a “road map.”
- **Move from general to specific in paragraphs.** Start each paragraph with a topic sentence that includes your main point, and then give supporting details. This approach always keeps your most important information at the beginnings of paragraphs, where readers tend to focus first while reading.

Another important consideration in organizing the document discussion is the way you handle facts versus opinions.

>> Informal Document Guideline 7: Separate Fact From Opinion

Some informal documents contain strong points of view. Others contain only subtle statements of opinion, if any. In either case, you must avoid any confusion about what constitutes fact or opinion. The safest approach in the document organization is to move logically from findings to your conclusions and, finally, to your recommendations. Because these terms are often confused, some working definitions are as follows:

- **Findings:** Facts you uncover (e.g., you observed severe cracks in the foundations of two adjacent homes in a subdivision).

- **Conclusions:** Summary of the document that emphasizes the information most important to your readers (e.g., you emphasize that foundation cracks occurred because the two homes were built on soft fill, where original soil had been replaced by construction scraps). Opinion is clearly a part of conclusions.
- **Recommendations:** Suggestions or action items based on your conclusions (e.g., you recommend that the foundation slab be supported by adding concrete posts beneath it). Recommendations are almost exclusively made up of opinions, but recommendations should clearly be grounded in the facts presented in the document.

>> Informal Document Guideline 8: Focus Attention in Your Conclusion

Letter and memo reports end with a section labeled *Findings*, *Conclusions*, or *Conclusions and Recommendations*, depending on whether the report is informative or analytical. (See Chapter 11 for more on the different purposes of reports.) Proposals use only the label *Conclusion* because they recommend actions throughout the document. (See Chapter 12 for more on proposals.) Choose the wording that best fits the content of your document. In all cases, this section gives details about your major findings, your conclusions, and, if called for, your recommendations. People often remember best what they read last, so think hard about what you place at the end of a document.

The precise amount of detail in your conclusion depends on which of these two options you choose for your particular document:

Option 1: If your major conclusions or recommendations have already been stated in the discussion, then you only need to restate them briefly to reinforce their importance (see Model 10–2, pp. 332–333).

Option 2: If the discussion leads up to, but has not covered, these conclusions or recommendations, then you may want to give more detail in this final section (see Model 10–1, pp. 330–331).

As in Model 10–1, lists are often mixed with paragraphs in the conclusion. Use such lists if you believe they will help readers remember your main points.

>> Informal Document Guideline 9: Use Attachments for Less Important Details

Informal documents are by definition short, yet detailed technical information is often needed for support. One solution to this dilemma is to place technical details in clearly labeled attachments that could include the following items:

- **Tables and figures:** Illustrations in informal documents usually appear in attachments unless it is crucial to include one within the text. Informal documents are so short that attached illustrations are easily accessible.
- **Costs:** It is best to list costs on a separate sheet. First, you do not want to bury important financial information within paragraphs. Second, readers must often circulate cost information, and a separate cost attachment is easy to photocopy and send.

>> Informal Document Guideline 10: Edit Carefully

Many readers judge you on how well you edit every document. A few spelling errors or some careless punctuation makes you seem unprofessional. Your career and your firm's future can depend on your ability to write final drafts carefully. Chapter 17 and the Handbook at the end of this text give detailed information about editing. For now, remember the following basic guidelines:

- Keep most sentences short and simple.
- Proofread several times for mechanical errors such as misspellings (particularly personal names).
- Triple-check all cost figures for accuracy.
- Make sure all attachments are included, are mentioned in the text, and are accurate.
- Check the format and wording of all headings and sub-headings.
- Ask a colleague to check over the document.

These guidelines help memo and letter reports and proposals accomplish their objectives. Remember, both your supervisors and your clients will judge you as much on communication skills as they do on technical ability. Consider each document part of your résumé. During your career, you will write many types of informal documents other than those presented here. If you grasp this chapter's principles, however, you can adapt to other formats.



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Informal Document Guidelines

- Plan well before you write
- Use letter or memo format
- Make text visually appealing
- Use the ABC format for organization
- Create the abstract as an introductory summary
- Put important details in the body
- Separate fact from opinion
- Focus attention in your conclusion
- Use attachments for less important details
- Edit carefully

>>> When to Use Formal Document Format

Like most people, you probably associate formal reports and proposals with important projects. What else justifies all that time and effort? In comparison to informal documents, formal documents usually (1) cover more complicated projects and (2) are longer than their informal counterparts. To prepare you to write effective formal documents, this section briefly describes four situations that would require formal documents. Second, this section provides guidelines for writing the main parts of a long document. Finally, this section discusses a complete long document from M-Global, Model 10–3 on pages 334–349.

Although complexity of subject matter and length are the main differences between formal and informal documents, sometimes there is another distinction: Formal reports and proposals may have a more diverse set of readers. In this case, readers who want just a quick overview can turn to the executive summary at the beginning or the conclusions

and recommendations at the end; technical readers who want to check your facts and figures can turn to discussion sections or appendixes; and all readers can flip to the table of contents for a quick outline of what sections the document contains. You must consider the needs of all these readers as you plan and write your formal documents.

The intended audience for formal reports and proposals can be internal or external, although the latter is more common for formal documents like these. Following are four situations at M-Global for which formal reports and proposals are appropriate:

- **Salary study and recommendations (internal):** Mary Kennelworth, a supervisor at M-Global's San Francisco office, has just completed a study of technicians' salaries among M-Global's competitors on the West Coast. What prompted the study was the problem she had hiring technicians to assist environmental engineers and geologists. Lately, some top applicants have been choosing other firms. Because the salary scales of her office are set by M-Global's corporate headquarters, she wants to give the main office some data showing that San Francisco starting salaries should be higher. Mary decides to submit a formal report, complete with data and recommendations for adjustments. Her main audience includes the San Francisco branch manager and Karrie Camp, Vice President for Human Resources in Baltimore (the company's top decision maker about salaries and other personnel matters).
- **Analysis of marketing problems (internal):** For several years, Jim Springer, Engineering Manager at M-Global's Houston office, has watched profits decline in onshore soils work. (In this type of work, engineers and technicians investigate the geologic and surface features of a construction site and then recommend foundation designs and construction practices.) One problem has been the "soft" construction market in parts of Texas. However, the slump in work has continued despite the recent surge in construction. In other words, some other company is getting the work. Jim and his staff have analyzed past marketing errors with a view to developing a new strategy for gaining new clients and winning back old ones. He plans to present the problem analysis and preliminary suggestions in a formal report. The main readers are his manager, the corporate marketing manager in Baltimore, and managers at other domestic offices who have positions that correspond to his.
- **Collapse of oil rig (external):** A 10-year-old rig in the North Sea recently collapsed during a mild storm. Several rig workers died, and several million dollars' worth of equipment was lost. Also, the accident created an oil spill that destroyed a significant amount of fish and wildlife before it was finally contained. M-Global's London office was hired to examine the cause of the collapse of this structure, which supposedly was able to withstand hurricanes. After three months of on-site analysis and laboratory work, M-Global's experts are ready to submit their report. It will be read by corporate managers of the firm, agencies of the Norwegian government, and members of several major wildlife organizations, and it will be used as the basis for some articles in magazines and newspapers throughout the world.
- **Sales proposal for work on wind turbine project (external):** A utility company in California plans to build 10 wind turbines in a desert valley in the southern part of the state. The "free" power that is generated will help offset the large increases

in fuel costs for the company's other plants. Although the firm has selected a turbine design and purchased the units, it must decide where to place them and what kind of foundations to use. Therefore it has sent out a Request For Proposal (RFP) to companies that have experience with foundation and environmental engineering. Louis Bergen, Engineering Manager at M-Global's San Francisco office, writes a proposal that offers to test the soils at the site, pinpoint the best locations for the heavy turbines, and design the most effective foundations.

As these four situations show, formal reports and proposals are among the most difficult on-the-job writing assignments you face in your career. Although some of these are written by a single author, most formal documents are written by teams of technical and professional people. In all cases, you must (1) understand your purpose, (2) grasp the needs of your readers, and (3) design a document that responds to these needs. The guidelines in the next two sections help you meet these goals.

>>> Strategy for Organizing Formal Documents

You will encounter different document formats in your career, depending on your profession and your specific employer. Whatever format you choose, however, there is a universal approach to good organization that always applies. This approach is based on these main principles, discussed in detail in Chapter 4:

Principle 1: Write different parts for different readers.

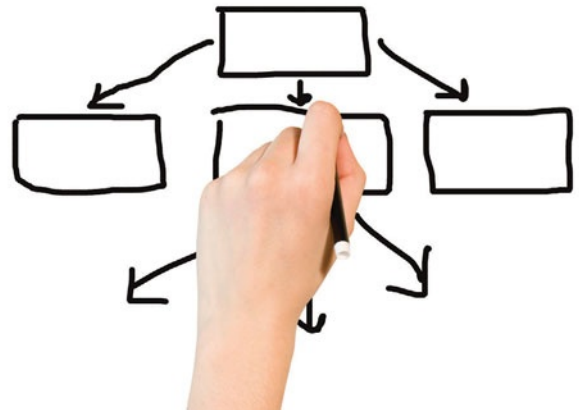
Principle 2: Place important information first.

Principle 3: Repeat key points when necessary.

These principles apply to long formal documents even more than they do to short documents, for the following reasons:

1. A formal document often has a very mixed audience—from laypersons to highly technical specialists to executives.
2. The majority of readers of formal documents focus on specific sections that interest them most, reading selectively each time they pick up the document.
3. Few readers have time to wade through a lot of introductory information before reaching the main point. They will get easily frustrated if you do not place important information first.

This chapter responds to these facts about readers of formal documents by following the ABC format (for *Abstract*, *Body*, *Conclusion*). As noted in Chapter 4, the three main rules are that you should (1) start with an abstract for decision makers, (2) put supporting



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ABC Format: Formal Document

■ ABSTRACT:

- Cover/title page
- Letter or memo of transmittal
- Table of contents
- List of illustrations
- Executive summary
- Introduction

■ BODY:

- Discussion sections
- [Appendixes—appear after text but support the body section]

■ CONCLUSION:

- Conclusions (for reports and proposals)
- Recommendations (for reports only)

details in the body, and (3) use the conclusion to produce action. This simple ABC format should be evident in all formal documents, despite their complexity. The particular sections of formal documents fit within the ABC format as shown on the right.

Several features of this structure deserve special mention. First, note that the generic abstract section includes five different parts of the document that help give readers a capsule version of the entire document. As we discuss shortly, the executive summary is by far the most important section for providing this *big picture* of the document. Second, appendixes are discussed within the body part of the outline, even though they are placed at the end of the document. The reason for this outline placement is that both appendixes and body sections provide supporting details for the document. Third, remember that the generic conclusion section in the ABC format can contain conclusions or recommendations or both, depending on the nature of the document.

Before moving to a discussion of the specific sections that make up the ABC format, take note of the use of main headings in complex formal documents. (See Figure 5–12, p. 137.) Much like chapter titles, these headings are often centered, in full caps, in bold type, and larger in font size than the rest of the text. They usually also begin on a new page. This way, each major section of the formal document seems to exist on its own. Subheadings are then used to indicate parts of the major sections.

Because formal documents may be longer and more complex than other forms of technical communication, it is important to help your readers navigate through the document. You may be used to thinking of navigation devices in Web pages and electronic documents, but they are also important for long print documents. For example, consider information that you can include in the header and footer of your document to help your reader find appropriate sections quickly. Your organization may require standardized information in headers and footers, such as the company name, the date of the document, or an identifying code. In very long documents, it may be useful to include section headings in the header, in the same way that this textbook includes chapter number and chapter title information in its headers. Obviously, including pagination in the header or footer of your document makes it much easier to find information. Many styles of pagination abound. Following are some guidelines for one commonly used pattern that is acceptable unless you have been instructed to use another:

- Use lowercase roman numerals for some or all of the front matter that precedes—and includes—the table of contents.
- Use Arabic numbers for items that follow the table of contents (all of which are listed in the table of contents).

- Continue the Arabic numbering for appendixes if they are relatively short. Long sets of appendixes sometimes have their own internal numbering (A–1, A–2, A–3 . . . ; B–1, B–2, B–3 . . .)

Dividers, colors on the edges of pages (*bleed indexes*), or tabbed sheets are also good ways to help readers find the document sections that they are interested in. Consider starting each section with a tabbed sheet so that the reader can “thumb” to it easily.

>>> Guidelines for the Nine Parts of Formal Documents

The nine parts of formal documents are as follows:

1. Cover/title page
2. Letter or memo of transmittal
3. Table of contents
4. List of illustrations
5. Executive summary
6. Introduction
7. Discussion sections
8. Conclusions and recommendations
9. End material

See Model 10–3 (pp. 334–349) for an example of these navigational elements and for an example of guidelines that follow.

Cover/Title Page

Formal documents are usually bound, often with a standard cover used for all documents in the writer’s organization. (Reports prepared for college courses, however, are often placed in a simple report cover.) Because the cover is the first item seen by the reader, it should be attractive and informative. It usually contains the same four pieces of information mentioned in the following list with regard to the title page; sometimes it has only one or two of these items.

Inside the cover is the title page, which should include the following four pieces of information:

- Project title (exactly as it appears on the letter/memo of transmittal)
- Your client’s or recipient’s name (“Prepared for . . .”)
- Your name and/or the name of your organization (“Prepared by . . .”)
- Date of submission

To make your title page or cover distinctive, you might want to place a simple illustration on it; however, do not clutter the page. Use a visual only if it reinforces a main point and if it can be done simply and tastefully. For example, assume that M-Global, Inc., submitted a formal report to a city in Georgia, reporting the results of a study of water pollution. The report writer decided to place the picture of a creek on the title page, punctuating the report's point about the water quality, as in Model 10–3 on page 334–349.

Letter/Memo of Transmittal

Letters or memos of transmittal are like an appetizer—they give the readers a taste of what is ahead. If your formal document is to readers outside your own organization, write a letter of transmittal. If it is to readers inside your organization, write a memo of transmittal. Figures 10–1 and 10–2 show examples of both. Use the following guidelines for constructing this part of your document:

>> Transmittal Guideline 1: Place the Letter/Memo Immediately after the Title Page

This placement means that the letter/memo is bound with the document, to keep it from becoming separated. Some organizations paper-clip this letter or memo to the front of the document or simply include it in the envelope with the document, making

MEMO

TO: Karrie Camp, Vice President for Human Resources
FROM : Abe Andrews, Personnel Assistant *aa*
SUBJECT: Report on Flextime Pilot Program at Boston Office
DATE: March 18, 2012

As you requested, I have examined the results of the six-month pilot program to introduce flextime to the Boston office. This report presents my data and conclusions about the use of flexible work schedules.

To determine the results of the pilot program, I asked all employees to complete a written survey. Then I followed up by interviewing every fifth person on an alphabetical list of office personnel. Overall, it appears that flextime has met with clear approval by employees at all levels. Productivity has increased and morale has soared. This report uses the survey and interview data to suggest why these results have occurred and where we might go from here.

I enjoyed working on this personnel study because of its potential impact on the way M-Global conducts business. Please give me a call if you would like additional details about the study.



M-Global Inc | 127 Rainbow Lane | Baltimore MD 21202 | 410.555.8175

■ Figure 10–1 ■ Memo of transmittal

12 Post Street
Houston Texas 77000
(713) 555-9781

Report #82-651
July 18, 2012

Belton Oil Corporation
PO Box 301
Huff Texas 77704

Attention: Mr. Paul A. Jones

**GEOTECHNICAL INVESTIGATION
DREDGE DISPOSAL AREA F
BELTON OIL REFINERY
HUFF, TEXAS**

This is the second volume of a three-volume report on our geotechnical investigation concerning dredge materials at your Huff refinery. This study was authorized by Term Contract No. 604 and Term Contract Release No. 20-6 dated May 6, 2012.

This report includes our findings and recommendations for Dredge Disposal Area F. Preliminary results were discussed with Mr. Jones on July 16, 2012. We consider the soil conditions at the site suitable for limited dike enlargements. However, we recommend that an embankment test section be constructed and monitored before dike design is finalized.

We appreciate the opportunity to work with you on this project, and we would like to thank Bob Berman and Cyndi Johnson for the help they provided on-site. We look forward to assisting you with the final design and providing materials-testing services.

Sincerely,



George H. Fursten
Geotechnical/Environmental Engineer
GHF/dnn



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■ **Figure 10-2** ■ Letter of transmittal

it a cover letter or memo. In so doing, however, they risk having it become separated from the document.

>> **Transmittal Guideline 2: Include a Major Point from Document**

Remember that readers are heavily influenced by what they read first in documents. Therefore, take advantage of the position of this section by including a major finding, conclusion, or recommendation from the document—besides supplying necessary transmittal information.

>> Transmittal Guideline 3: Acknowledge Those Who Helped You

Recognizing those who have been particularly helpful with your project gives them recognition and identifies you as a team player. It reflects well on you and on your organization. Figure 10–2 includes a thank you to members of the client’s organization.

>> Transmittal Guideline 4: Follow Letter and Memo Conventions

Like other letters and memos, letters and memos of transmittal should be easy to read, inviting readers into the rest of the document. Keep introductory and concluding paragraphs relatively short—no more than three to five lines each. Also, write in a conversational style, free of technical jargon and stuffy phrases such as “per your request” or “enclosed herewith.” See the models at the end of Chapter 6 for more details concerning letter/memo format. For now, here are some highlights about the mechanics of format:

Letters and Memos

- Use single spacing and ragged-right-edge copy, even if the rest of the document is double-spaced and fully justified.
- Use only one page.

Letters

- Include company project number with the letter date.
- Spell the reader’s name correctly.
- Be sure the inside address includes the mailing address to appear on the envelope.
- Use the reader’s last name (“Dear Mr. Jamison:”) in the salutation or attention line because of the formality of the document—unless your close association with the reader would make it more appropriate to use first names (“Dear Bill:”).
- Usually include a project title. It is treated like a main heading. Use concise wording that matches wording on the title page.

- Use “Sincerely” as your closing.
- Include a line to indicate those who will receive copies of the document (“cc” or just “c” or “copy” for copy, “pc” for photocopy).

Memos

- Give a clear description of the project in the subject line of the memo, including a project number if there is one.
- Include a distribution list to indicate those who will receive copies.

Transmittal Correspondence Guidelines

- Place the letter/memo immediately after the title page
- Include a major point from document
- Acknowledge those who helped you
- Follow letter and memo conventions

Table of Contents

Your contents page acts as an outline. Many readers go there right away to grasp the structure of the document and then return repeatedly to locate document sections of most interest to them. Most word-processing programs allow you to generate a table of contents automatically from tagged headings, but tables of contents generated this way

must often be edited, especially if you have decided to leave out lower-level headings (see Table of Contents Guideline 3). Guidelines follow for assembling this important component of your document; see page 334–349 in Model 10–3 for an example.

>> Table of Contents Guideline 1: Make It Very Readable

The table of contents must be pleasing to the eye so that readers can find sections quickly and see their relationship to each other. Be sure to

- Space items well on the page
- Use indenting to draw attention to subheadings
- Include page numbers for every heading and subheading, unless there are many headings in a relatively short document, in which case you can delete page numbers for all of the lowest-level headings listed in the table of contents

>> Table of Contents Guideline 2: Use the Contents Page to Reveal Document Emphases

Choose the wording of headings and subheadings with care. Be specific yet concise so that each heading listed in the table of contents gives the reader a good indication of what the section contains.

Readers associate the importance of document sections with the number of headings and subheadings listed in the table of contents. If, for example, a discussion section called “Description of the Problem” contains many more heading breakdowns than other sections, you are telling the reader that the section is more important. When possible, it is best to have about the same number of breakdowns for document sections of about the same importance. In short, the table of contents should be balanced.

>> Table of Contents Guideline 3: Consider Leaving Out Low-Level Headings

In very long documents, you may want to unclutter the table of contents by removing lower-level headings. As always, the needs of the readers are the most important criterion to use in making this decision. If you think readers need access to all levels of headings on the contents page, keep these headings there. If you think they would prefer a simple contents page instead of a comprehensive one, delete all the lowest-level headings from the table of contents.

>> Table of Contents Guideline 4: List Appendixes

Appendixes include items such as tables of data or descriptions of procedures that are inserted at the end of the document. Typically, they are listed at the end of the table of contents. They may be paged with Arabic numerals, like the rest of the document. However, sometimes no page numbers are given in the table of contents, because many appendixes contain *off-the-shelf* material such as résumés or project sheets and are thus individually paged (e.g., Appendix A might be paged A–1, A–2, A–3, etc.). Tabs on the edges of pages can help the reader locate these sections.

>> Table of Contents Guideline 5: Use Parallel Form in All Entries

All headings in one section, and sometimes even all headings and subheadings in the document, have parallel grammatical form. Readers find mixed forms distracting. For example, “Subgrade Preparation” and “Fill Placement” are parallel because they are both the same type of phrase. However, if you switch the wording of the first item to “Preparing the Subgrade” or “How to Prepare the Subgrade,” parallel structure is lost.

Table of Contents Guidelines

- Make it very readable
- Use the contents page to reveal document emphases
- Consider leaving out low-level headings
- List appendixes
- Use parallel form in all entries
- Proofread carefully

>> Table of Contents Guideline 6: Proofread Carefully

The table of contents is one of the last document sections to be assembled; therefore it often contains errors. Wrong page numbers and incorrect headings are two common mistakes. Another is the failure to show the correct relationship of headings and subheadings. If you have used your word-processing software to generate a table of contents automatically, you may need to generate it again, after you have completed your final version of the document. Obviously, errors in the table of contents can confuse the reader and prove embarrassing to the writer. Proofread this section carefully.

List of Illustrations

Illustrations within the body of the document are usually listed on a separate page right after the table of contents. When there are few illustrations, another option is to list them at the bottom of the table of contents rather than on a separate page. In either case, this list should include the number, title, and page number of every table and figure within the body of the document. If there are many illustrations, separate the list into tables and figures. See the example on page 334–349 in Model 10–3. (For more information on illustrations, see Chapter 13.)

Executive Summary

No formal document would be complete without an executive summary. This short section provides decision makers with a capsule version of the document. Consider it a stand-alone section that should be free of technical jargon. In some cases, a copy of the executive summary may be circulated and filed separate from the document (see Figure 10–3). Follow these basic guidelines in preparing this important section of your formal documents:

>> Executive Summary Guideline 1: Put It on One Page

The best reason to hold the summary to one page is that most readers expect and prefer this length. It is a comfort to know that somewhere within a long document there is one page to which one can turn for an easy-to-read overview. Moreover, a one-page length permits easy distribution at meetings. When the executive summary begins to crowd your page, it

EXECUTIVE SUMMARY

Quarterly monitoring of groundwater showed the presence of nickel in Well M-17 at the Hennessey Electric facility in Jones, Georgia. Nickel was not detected in any other wells on the site. Hennessey retained M-Global's environmental group to determine the source of the nickel.

The project consisted of four main parts. First, we collected and tested 20 soil samples within a 50-yard radius of the well. Second, we collected groundwater samples from the well itself. Third, we removed the stainless steel well screen and casing and submitted them for metallurgical analysis. Finally, we installed a replacement screen and casing built with Teflon.

The findings from this project are as follows:

- The soil samples contained *no* nickel.
- We found *significant* corrosion and pitting in the stainless steel screen and casing that we removed.
- We detected *no* nickel in water samples retrieved from the well after replacement of the screen and casing.

Our study concluded that the source of the nickel in the groundwater was corrosion on the stainless steel casing and screen.

■ **Figure 10-3** ■ Executive summary—formal report

is acceptable to switch to single-spacing if such a change helps keep the summary on one page—even though the rest of the document may be space-and-a-half or double-spaced.

Some extremely long formal documents may require that you write an executive summary of several pages or longer. In this case, you must still provide the reader with a section that summarizes the document in less than a page. The answer to this dilemma is to write a brief *abstract*, a condensed version of the executive summary directed to the highest-level decision makers. The abstract should be placed right before the executive summary. (See Chapter 9 for further discussion of abstracts.)

>> Executive Summary Guideline 2: Avoid Technical Jargon

Include only that level of technical language the decision makers comprehend. It makes no sense to talk over the heads of the most important readers.

>> Executive Summary Guideline 3: Include Only the Important Conclusions and Recommendations

The executive summary mentions only the major points of the document. An exhaustive list of appropriate findings, conclusions, and recommendations can come later, at the end of the document. If you have trouble deciding what is most important, put yourself in the position of the readers. What information is most essential for them? If you want to leave

them with one, two, or three points about the document, what would these points be? That is the information that belongs in the executive summary.

>> **Executive Summary Guideline 4: Avoid References to the Document Body**

Avoid the tendency to say that the document provides additional information. It is understood that the executive summary is only a generalized account of the document's contents. References to later sections do not provide the busy reader with further understanding.

An exception is those instances when you are discussing issues that involve danger or liability, when it may be necessary to add qualifiers in your summary—for example, “As noted in this document, further study will be necessary.” Such statements protect you and the client in the event the executive summary is removed from the document and used as a stand-alone document.

>> **Executive Summary Guideline 5: Use Paragraph Format**

Whereas lists are often appropriate for body sections of a document, they can give executive summaries a fragmented effect. Instead, the best summaries create unity with a series of relatively short paragraphs that flow together well. Within a paragraph, there can be a short listing of a few points for emphasis (see Figure 10–3 on page 334–349), but the listing should not be the main structural element of the summary.

Occasionally, you may be convinced that the paragraph approach is not desirable. For example, a project may involve a series of isolated topics that do not mesh into unified paragraphs. In this case, use a modified list. Start the summary with a brief introductory paragraph, followed by a numbered list of three to nine points. Each numbered point should include a brief explanation. For example,

1. Sewer Construction: We believe that seepage influx can be controlled by . . .
2. Geologic Fault Evaluation: We found no evidence of surficial . . .

Executive Summary Guidelines

- Put it on one page
- Avoid technical jargon
- Include only the important conclusions and recommendations
- Avoid references to the document body
- Use paragraph format
- Write the executive summary last

>> **Executive Summary Guideline 6: Write the Executive Summary Last**

Only after finishing the document do you have the perspective to write a summary. Approach the task in a logical manner. First, sit back and review the document from beginning to end, and then ask yourself, “What would my readers really need to know if they had only a minute or two to read?” The answer to that question becomes the core of your executive summary.

Introduction

View this section as your chance to prepare both technical and nontechnical readers for the discussion ahead. You do not need to summarize the document, because your executive

summary has accomplished that goal. Instead, give information on the document's purpose, scope, and format, as well as a project description. Follow these basic guidelines, as reflected on page 334–349 in Model 10–3.

>> **Introduction Guideline 1: State Your Purpose and Lead into Subsections**

The purpose statement for the document should appear immediately after the main introduction heading (e.g., “This document presents M-Global’s foundation design recommendations for the new Hilltop Building in Franklin, Maine”). Follow it with a sentence that mentions the introduction subdivisions to follow (e.g., “This introduction provides a description of the project site and explains the scope of activities we conducted”).

>> **Introduction Guideline 2: Include a Project Description**

Here you must be precise about the project. Depending on the type of project, you may be describing a physical setting, a set of problems that prompted the document study, or some other data. The information may have been provided to you, or you may have collected it yourself. Accuracy in this section helps prevent any later misunderstandings between you and the reader. (When the project description is too long for the introduction, sometimes it is placed in the body of the document.)

>> **Introduction Guideline 3: Include Scope Information**

This section outlines the precise objectives of the project. Include all necessary details, using bulleted or numbered lists when appropriate. Your listing or description should parallel the order of the information presented in the body of the document. Like the project description, this subsection must be accurate in every detail. Careful and thorough writing here can prevent later misunderstandings about the tasks you were hired to perform.

>> **Introduction Guideline 4: Consider Including Information on Document Organization**

Often, the scope section lists information as it is presented in the document. If this is not the case, end the introduction with a short subsection on the document organization where you can give readers a brief preview of the main sections that follow. In effect, the section acts as a condensed table of contents and may list the document's major sections and appendixes.

Discussion Sections

Discussion sections make up the longest part of formal documents. In general, they are written for the most technically oriented members of your audience. You can focus on facts and opinions, demonstrating the technical expertise that the reader expects from

Introduction Guidelines

- State your purpose and lead into subsections
- Include a project description
- Include scope information
- Consider including information on document organization

you. Because the discussion sections will be the longest part of your document, they should be organized carefully. Common patterns of organization, such as a problem solution or chronological structure, can be used to organize the entire body of the document, or they may be used to organize specific sections. (See Chapter 4 for more about the common patterns of organization.) The discussion sections often include technical definitions, descriptions, and process explanations. They may also include information about budgets, schedules, and other resources. General guidelines for writing the document discussion are listed next. Chapter 11 and Chapter 12 list guidelines for discussion sections for specific types of documents. For a complete example of the discussion component, see the formal document example in Model 10–3 on pages 334–349.

>> Discussion Guideline 1: Move from Facts to Opinions

As you have learned, the ABC format requires that you start your formal document with a summary of the most important information—that is, you skip right to essential conclusions and recommendations the reader needs and wants to know. Once into the discussion section, however, you back up and adopt a strategy that parallels the stages of the technical project itself. You begin with hard data and move toward conclusions and recommendations (i.e., those parts that involve more opinion). There are two reasons for this approach, one ethical and the other practical. First, as a professional, you are obligated to draw clear distinctions between what you observe and what you conclude or recommend. Second, documents are usually more persuasive if you give readers the chance to draw conclusions for themselves.

>> Discussion Guideline 2: Use Frequent Headings and Subheadings

Headings give readers handles by which to grasp the content of your document. They are especially needed in the document body, which presents technical details. Your readers view headings, collectively, as a sort of outline by which they can make their way easily through the document.

>> Discussion Guideline 3: Use Listings to Break Up Long Paragraphs

Long paragraphs full of technical details irritate readers. Use paragraphs for brief explanations, not for descriptions of processes or other details that could be listed.

>> Discussion Guideline 4: Use Illustrations for Clarification and Persuasion

A simple table or figure can sometimes be just the right complement to a technical discussion in the text. Incorporate illustrations into the document body to make technical information accessible and easier to digest.

>> Discussion Guideline 5: Place Extra Detail in Appendixes

Today's trend is to place cumbersome detail in appendixes that are attached to formal documents, rather than weighing down the discussion with this detail. In other words, you give readers access to supporting information without cluttering up the text of the

formal report or proposal. Of course, you must refer to the appendixes in the body of the document and label appendixes clearly so that readers can locate them easily.

Conclusions and Recommendations

This section of the document gives readers a place to turn to for a comprehensive description—sometimes in the form of a listing—of all conclusions and recommendations. The points may or may not have been mentioned in the body of the document, depending on the length and complexity of the document. It can sometimes be difficult to decide whether to use the term *conclusions* or the term *recommendations*, or both. Your organization may have guidelines for how to label the last section of formal documents; however, these general guidelines can help you decide which term best describes the final section of your document. *Conclusions*, on the one hand, summarize the content of your document. They emphasize the information that you feel is most important for your reader. *Recommendations*, on the other hand, are actions you are suggesting based on your conclusions. For example, your conclusion may be that there is a dangerous level of toxic chemicals in a town's water supply, and your recommendation may be that the toxic site near the reservoir should be cleaned immediately. As noted earlier, the final section of proposals is usually labeled *Conclusions*.

What distinguishes this final section of the document text from the executive summary is the level of detail and the audience. The section on conclusions and recommendations provides an exhaustive list of conclusions and recommendations for technical and management readers. The executive summary provides a selected list or description of the most important conclusions and recommendations for decision makers, who may not have technical knowledge.

In other words, view the section on conclusions and recommendations as an expanded version of the executive summary. It usually assumes one of these three headings, depending, of course, on the content:

1. Conclusions
2. Recommendations
3. Conclusions and Recommendations

Another option for documents that contain many conclusions and recommendations is to separate this last section into two sections: (1) "Conclusions" and (2) "Recommendations."

End Material

One kind of end material—appendixes—is mentioned in the context of the discussion section. Note that formal documents may also contain works-cited pages or bibliographies, which should be included in the end materials. See Chapter 9 and, at the end of this book, the Handbook for guidelines on formatting in-text citations and bibliography entries. Finally, very long documents may include indexes.

Discussion Guidelines

- Move from facts to opinions
- Use frequent headings and subheadings
- Use listings to break up long paragraphs
- Use illustrations for clarification and persuasion
- Place extra detail in appendixes

>>> Formal Report Example

Model 10–3 (pp. 334–349) provides a long and formal technical report from M-Global, Inc. It contains the main sections discussed previously, including the list of illustrations. Marginal annotations indicate how the model reflects proper use of this chapter’s guidelines for format and organization.

The report results from a study that M-Global completed for the city of Winslow, Georgia. Members of the audience come from both technical and nontechnical backgrounds. Some are full-time professionals hired by the city, whereas others are part-time, unpaid citizens appointed by the mayor to explore environmental problems. The paid professionals include engineers, environmental specialists, accountants, city planners, managers, lawyers, real estate experts, and public relations specialists. The part-time appointees include citizens who work in a variety of blue-collar and white-collar professions or who are homemakers.

>>> Chapter Summary

- Complex documents like reports and proposals may be formatted as informal documents or as formal documents.
- Usually, the choice of format is obvious, or it is assigned, but sometimes the reader must choose which format to use based on how narrow or broad the scope is and how large and diverse the audience is.
- Informal documents are formatted as memos, for communication within organizations, or as letters, for communications between organizations.
- Informal documents generally have a narrow, specific focus.
- Informal documents are usually written for one reader or a few readers with similar backgrounds.
- The ABC format can help writers organize informal documents.
- Less important details should be attached to informal documents as appendixes.
- Formal documents cover topics that are more complex or projects with larger scope.
- Formal documents are usually written for multiple readers with different technical expertise.
- Formal documents have up to nine main parts: a cover and/or title page, a letter or memo of transmittal, a table of contents, a list of illustrations, an executive summary, an introduction, a discussion, conclusions and/or recommendations, and end material.
- Because of their length and complexity, formal documents need navigation elements such as headings and subheadings, as well as clear organization.
- Informal and formal documents should be visually appealing and carefully edited.

>>> Learning Portfolio

Communication Challenge The Ethics of Clients Reviewing Report Drafts

Last week, Hank Wallace of M-Global's Kenya office completed the draft of an ocean exploration project for the Republic of Cameroon (see the second project sheet on page 473 in Model 12–6). As is routine with major reports, Hank showed the client a draft before the final draft was submitted. For the first time in his career, he was asked by the client to make changes he thinks are difficult to justify by project data. This case study is an explanation of why M-Global shares report drafts with some clients, as well as some background on the ocean exploration project. It ends with questions and comments for discussion and an assignment for a written response to the Challenge.

Sharing Drafts with Clients

In M-Global's business, some of the firm's reports must be submitted both to the paying client and to regulatory agencies of the government. This dual audience has created a review procedure common in the industry. Client firms have an opportunity to review a draft and make suggestions before both they and the regulatory agencies are sent final drafts.

For example, a U.S. mining company hired M-Global to examine a Siberian site to determine if gold reserves could be mined without damaging the delicate permafrost surface of the tundra. Because the Russian government regulates development of the region, it received a final copy of the report. However, before the final copy was submitted to the government, M-Global shared a draft with engineers and executives from the mining company. These client representatives questioned several technical assumptions M-Global made about the site, but M-Global had adequate justifications for its work. In the end, M-Global made no change in its original draft recommendation—that is, that further study was needed before mining was permitted in the permafrost region.

In another case, however, a client's review of a report on a dam in the midwestern United States prompted M-Global to adjust its report before submission of the final draft to the state's Department of Natural Resources, which regulates high-hazard dams. The owners of the dam—who paid for the study—convinced M-Global that the report should emphasize the fact that poor installation of a guard-rail over the dam created a drainage problem. When heavy rains came, soil washed out an embankment near the dam's spillway. The first draft had failed to mention that the state's transportation group bore some responsibility for the dam's problems.

The Ocean Exploration Report Review

The draft review of the ocean exploration report for the Republic of Cameroon did not go as smoothly as the two reviews just described. Major differences of opinion were evident between the M-Global project manager and the client.

As indicated on the project sheet on page 473 in Model 12–6, M-Global engineers developed conclusions and recommendations for the Cameroon coastal site. For the most part, they found that the offshore environment where they did the study would be too environmentally sensitive to drill offshore wells or run pipelines. There were two locations where a pipeline might be placed safely, but even in this case, some environmental damage was likely. When the Ministry of Mines and Energy got the draft report, the client asked for a meeting.

At the meeting the following week, M-Global engineers reviewed their findings, conclusions, and recommendations with the client. Ultimately, M-Global managers were asked to change the wording in the report to present a more favorable view of oil exploration at the site because, in the client's opinion, M-Global was being too conservative in its conclusions. If M-Global would just adjust some wording so as not to emphasize what was, after all, only possible environmental damage, then the Cameroon government might be provided the support it needed to develop this potentially rich oil field. Cameroon, the client argued, needed oil revenues to improve its economy and assist poor farmers with the transition to a modern economy. M-Global was not being asked to alter the facts—only to adjust the tone of the language.

Back at the office, the M-Global project manager met with the branch manager and later with corporate staff via teleconference. The project manager presented the facts of the project and a summary of the meeting. To all present, it was clear that the relationship with a long-term client was at stake.

Questions and Comments for Discussion

1. How should M-Global, Inc., respond to the client's request?
2. Generally, do you think M-Global's procedure for reviewing report drafts with clients is ethically sound? Support your answer.
3. If you answered yes to Question 2, do you have any suggestions to improve the procedure for this client report review? In other words, how might the process be adjusted to reduce the potential for misunderstandings and abuse?

4. If you answered no to Question 2, is there any circumstance in which you would support the review of a report draft by a client before final submission to the client and its regulatory agency?
5. It is often said, in this text and elsewhere, that collaboration is essential in the workplace. Describe the kinds of on-the-job situations where you think collaboration between writer and reader would be useful, appropriate, and ethical.

Write About It

Assume the role of the project manager in this Communication Challenge. In preparing for your teleconference meeting, you have been thinking about whether M-Global's procedure for reviewing report drafts with clients is ethical (see Question 2). Write a memo to Erik Schell, Vice President of International Operations, explaining your opinion. Respond to the issues raised in Question 3 or Question 4.

Collaboration at Work Suggestions for High School Students

General Instructions

Each Collaboration at Work exercise applies strategies for working in teams to chapter topics. The exercise assumes you (1) have been divided into teams of about three to six students, (2) will use time inside or outside of class to complete the case, and (3) will produce an oral or written response. For guidelines about writing in teams, refer to Chapter 3.

Background for Assignment

This chapter introduced you to two basic formats for reports—informal memo and letter formats and the formal format that includes a cover, letter or memo of transmittal, and front and back material that are not part of the informal document format. As with other collaborative efforts, first, you share ideas in a nonjudgmental way; then you choose what should be included in the report and how the report should be formatted and organized based on your team discussions.

Team Assignment

Assume that an association of colleges and universities has asked your team to help write a short report to be sent to high school students. The report's purpose is to assist students in selecting a college or university. First, your team will decide if the report should be formatted as an informal document or as a formal document. Then your team will prepare an outline for the body of the report by (1) choosing several headings that classify groupings of recommendations and (2) providing specific recommendations within each grouping. For example, one grouping might be "Support for Job Placement," with one recommendation in this grouping being "Request data on the job placement rates of graduates of the institution." After deciding on your format and producing your outline, share the results with other teams in the class. Be prepared to discuss why you have chosen the informal or formal format and what organization principles you used in planning the body of the report. (See Chapter 4 for more on organization principles.)

Assignments

Assignments can be completed either as individual exercises or as team projects, depending on the directions of your instructor. Your instructor will ask you to prepare a response that can be delivered as an oral presentation for discussion in class. Analyze the context of each Assignment by considering what you learned in Chapter 1 about the context of technical writing, and answer the following questions:

- What is the purpose of the document to be written?
- What result do you hope to achieve by writing it?
- Who are your readers and what do they want from your document?
- What method of organization is most useful?

1. Analysis: Informal documents

Use Model 10-1 and Model 10-2 on pages 330-333, for this assignment. Each of these reports is addressed to one reader, but the readers of these reports are quite different. For each report

- Identify the likely audience for the document.
- Discuss how you identified the characteristics of the audience for each document.
- Compare the two documents. How is each document organized to answer a question or solve a problem for the reader?
- Does the writer of each document present a professional image? Explain.

2. Analysis: Formal Report

Locate a formal report written by a private firm or government agency, or use a long report provided by your instructor. (You can use the advanced search tools in an Internet search engine to find a report that is in PDF format.) Determine the degree to which the example follows the guidelines in this chapter. Depending on the instructions given by your teacher, choose between the following options:

- Present your findings orally or in writing.
- Select part of the report or all of the report.

3. Analysis: Executive Summary

The 9/11 Commission Report was widely praised for its excellent writing, especially for its appropriateness to an audience that included a general, international public, as well as people in government agencies. It was even nominated

for a National Book Award for nonfiction. Find the executive summary for the 9/11 Commission Report (located at http://www.9-11commission.gov/report/911Report_Exec.pdf). Evaluate it as a stand-alone document. What techniques did the writers use to prepare the executive summary for the media and general public who would be reading it? Can you explain why the executive summary does not follow all of the guidelines in this chapter?

Depending on the instructions given by your teacher, present your findings in writing or be prepared to discuss them in class.

4. Analysis: Introduction

Review the chapter guidelines for writing an effective introduction to a formal report. Then evaluate the degree to which the following example follows or does not follow the guidelines presented.

INTRODUCTION

M-Global, Inc., has completed a three-week study of the manufacturing and servicing processes at King Radio Company. As requested, we have developed a blueprint for ways in which computer-aided testing (CAT) can be used to improve the company's productivity and quality.

Project Description

Mr. Dan Mahoney familiarized our project team with the problems that prompted this study of CAT. According to Mr. Mahoney, the main areas of concern are as follows:

- Too many units on the production line are failing postproduction testing and thus returning to the repair line.
- Production bottlenecks are occurring throughout the plant because of the testing difficulties.
- Technicians in the servicing center are having trouble repairing faulty units because of their complexity.
- Customers' complaints have been increasing, about both new units under warranty and repaired units.

Scope

From May 3 through May 5, 2012, M-Global, Inc., had a three-person team of experts working at the King Radio Company plant. This team interviewed many personnel, observed all the production processes, and acquired data needed to develop recommendations. On returning to the M-Global office, team members met to share their observations and develop the master plan included in this report.

Report Format

This report is organized primarily around the two ways that CAT can improve operation at the King Radio Company plant. Based on the detailed examination of the plant's problems in this regard, the report covers two areas for improvement and ends with a section that lists main conclusions and recommendations. The main report sections are as follows:

- Production and Servicing Problems at King Radio
- CAT and the Manufacturing Process
- CAT and the Servicing Process
- Major Conclusions and Recommendations

The report ends with two appendixes. Appendix A offers detailed information on several pieces of equipment we recommend that you purchase. Appendix B provides three recent articles from the journal *CAT Today*, all of which deal with the application of CAT to production and service problems similar to those you are experiencing.

Practice Assignments

Follow these general guidelines for the practice assignments:

- Print or design a letterhead when necessary.
- Use whatever letter, memo, or e-mail format your instructor requires.
- Invent addresses when necessary.
- Invent any extra information you may need for the correspondence, but do not change the information presented here.

5. Practice: Informal Report Based on Internet Search

Use the Internet to collect actual information, or a list of sources that may contain information, about a topic that relates to your academic major. Then write an informal report in which you analyze (1) the ease with which the Internet allowed you to collect information on your topic and (2) the quality of the sources or information you received. Your audience is your instructor, who will let you know the degree of knowledge you can assume he or she has on this topic.

6. Practice, M-Global Context: Memo Report

Assume you are an M-Global field engineer working at the construction site of a nuclear power plant in Jentsen, Missouri. For the past three weeks, your job has been to observe the construction of a water-cooling tower, a large cylindrical structure. As consultants to the plant's construction firm, you and your M-Global crew were hired to make sure that work proceeds properly and on schedule. As the field engineer, you are supposed to report any problems in writing to your project manager, John Raines, back at your St. Louis office. Then he will contact the construction firm's office, if necessary.

Write a short memo report to Raines. Take the following randomly organized information and present it in a clear, well-organized fashion. If you wish, add information of your own that might fit the context.

- Three cement pourings for the tower wall were delayed an hour each on April 21 because of light rain.
- Cement-truck drivers must slow down while driving through the site. Other workers complain about the excessive dust raised by the trucks.
- Mary Powell, an M-Global safety inspector on the crew, cited 12 workers for not wearing their hard hats.
- You just heard from one subcontractor, Allis Wire, Inc., that there will be a two-day delay in delivering some steel reinforcing wires that go into the concrete walls. That delay will throw off next week's schedule. Last Monday's hard rain and flooding kept everyone home that day.
- It is probably time once again to get all the subcontractors together to discuss safety at the tower site. Recently, two field hands had bad cuts from machinery.
- Although there have not been any major thefts at the site, some miscellaneous boards and masonry pieces are missing each day—probably because nearby residents (doing small home projects) think that whatever they find at the site has been discarded. Are additional “No trespassing” signs needed?
- Construction is only two days behind schedule, despite the problems that have occurred.

7. Practice, M-Global Context: Letter Report

This project requires some research. Assume that your college plans either to embark on a major recycling effort or to expand a recycling program that has already started. Put yourself in the role of an M-Global environmental scientist or technician who has been asked to recommend these recycling changes.

First, do some research about recycling programs that have worked in other organizations. A good place to start is a periodical database such as EBSCOhost or J-Stor, which will lead you to some magazine articles of interest. Choose to discuss one or more recoverable resources, such as paper, aluminum, cardboard, plastic, or glass bottles. Be specific about how your recommendations can be implemented by the organization or audience for which you are writing.

8. Practice: Research-Based Formal Report

Complete the following procedure for writing a research-based report:

- Use library and Internet resources to research a general topic in a field that interests you. Do some preliminary reading to screen possible specific topics.
- Choose three to five specific topics that require further research and for which you can locate information.
- Work with your instructor to select the one topic that best fits this assignment, given your interests and the criteria set forth here.
- Develop a simulated context for the report topic, whereby you select a purpose for the report, a specific audience to whom it could be addressed (as if it were a real report), and a specific role for you as a writer.

For example, assume you have selected “Earth-Sheltered Homes” as your topic. You might be writing a report to the manager of a local design firm on the features and construction techniques of such structures. As a newly hired engineer or designer, you are presenting information so that your manager can decide whether the firm might want to begin building and marketing such homes. This report might present only data, or it could present data and recommendations.

- Write the report according to the format guidelines in this chapter and in consideration of the specific context you have chosen.
- Document your sources appropriately (see Chapter 9).

9. Practice: Work-Based Formal Report

This assignment is based on the work experience that you may have had in the past or that you may be experiencing now.

- Choose five report topics that are based on your current or past work experience. For example, you could choose “warehouse design” if you stock parts, “checkout procedure” if you work behind the counter at any retail store, “report production procedures” if you work as a secretary at an engineering firm, and so on. In other words, find a subject that you know about, or about which you can find more information, especially through interviews.
- Work with your instructor to select the one topic that holds out the best possibilities for a successful report on the basis of the criteria given here.
- Develop a context for the report in which you give yourself a role in the company where you work(ed). This role should be one in which you would actually write a formal in-house or external report about the topic you have chosen, but the role does not have to be the exact one you had or have. Then select a precise purpose for which you might be writing the report, and finally, choose a set of readers who might read such a report within or outside the organization. Your report can be a presentation of data and conclusions or a presentation of data, conclusions, and recommendations.
- Follow the guidelines included in this chapter for format and organization.

10. Practice, M-Global Context: Formal Report

For this assignment, place yourself in a role of your choosing at M-Global, Inc. Use the following procedure, which may be modified by your instructor:

- Review pages 310–311, which list M-Global cases for formal reports to get a sense of when formal reports are used at companies like M-Global.
- Review the M-Global information in Model 1–1 on pages 25–34, especially with regard to the kinds of jobs people hold at the company and the kinds of projects that are undertaken.
- Choose a specific job that you could assume at M-Global, based on your academic background, your work experience, or your career interests.
- Choose a specific project that (a) could conceivably be completed at M-Global by someone in the role you have

chosen, (b) would result in a formal report directed either inside or outside the company, and (c) would be addressed to a complex audience at two or three of the levels indicated on the Planning Form at the end of the book.

- Be sure you have access to information that will be used in this simulated report—for example, from work experience, from a term paper or class project in another course, or from your interviews of individuals already in the field. (For this assignment, you may want to talk with a professional, such as a recent graduate in your major.)
- Prepare a copy of the Planning Form at the end of the book for your instructor’s approval before proceeding further with the project.
- Complete the formal report, following the guidelines in this chapter.



11. Ethics Assignment

Illustrations on cover pages of formal reports are one strategy for attracting the readers’ attention to the document. Note the use of an illustration in Model 10–3, along with the rationale on page 341. Do you think Model 10–3 uses its graphics in an ethically sound way to engage the reader with the report? Why or why not? How do you determine whether a cover page illustration is an appropriate persuasive tool, on the one hand, or an inappropriate attempt to manipulate the reader, on the other? Give hypothetical examples, or find examples from reports available on the Internet.



12. Informal Report: International Context

Investigate features such as style, format, structure, and organization of formal reports written in another country. For this assignment, it would be best to interview someone who does business in another country and, if possible, to get an actual report that you can submit. Write a memo report to your instructor that presents the results of your study.



13. A.C.T. N.O.W. Assignment (Applying Communication To Nurture Our World)

Interview a member of the campus staff responsible for adopting energy-saving measures at your college or university. Focus on one or more technical or social strategies, such as alternative-energy vehicles, new technology for regulating energy systems, advanced insulation, variable work hours, and modification of human behaviors related to energy use. Then write a short report on the relative success of the strategies you have researched. If the report is well reviewed by your instructor, consider seeking wider distribution of the report by submitting it (or a version of it) to the campus—assuming you have the permission of the person interviewed.

12 Post Street
Houston Texas 77000
(713) 555-9781

April 22, 2012

Big Muddy Oil Company Inc
12 Rankin St
Abilene TX 79224

ATTENTION: Mr. James Smith, Engineering Manager

**SHARK PASS STUDY
BLOCK 15, AREA 43-B
GULF OF MEXICO**

Includes specific title.

Uses optional heading
for abstract part of
ABC format.

INTRODUCTORY SUMMARY

You recently asked our firm to complete a preliminary soils investigation at an off-shore rig site. This report presents the tentative results of our study, including major conclusions and recommendations. A longer, formal report will follow at the end of the project.

Draws attention to *main*
point of report.

On the basis of what we have learned so far, it is our opinion that you can safely place an oil platform at the Shark Pass site. To limit the chance of a rig leg punching into the seafloor, however, we suggest you follow the recommendations in this report.

Gives on-site details of
project—dates, location,
tasks.

WORK AT THE PROJECT SITE

On April 15 and 16, 2012, M-Global's engineers and technicians worked at the Block 15 site in the Shark Pass region of the gulf. Using M-Global's leased drill ship, *Seeker II*, as a base of operations, our crew performed these main tasks:

- Seismic survey of the project study area
- Two soil borings of 40 feet each

Uses *lead-in* to subsec-
tions that follow.

Both seismic data and soil samples were brought to our Houston office the next day for laboratory analysis.

LABORATORY ANALYSIS

On April 17 and 18, our lab staff examined the soil samples, completed bearing capacity tests, and evaluated seismic data. Here are the results of that analysis.

Soil Layers

Highlights most impor-
tant point about soil
layer—that is, the *weak*
clay.

Our initial evaluation of the soil samples reveals a 7- to 9-foot layer of weak clay starting a few feet below the seafloor. Other than that layer, the composition of the soils seems fairly typical of other sites nearby.



M-Global Inc | 127 Rainbow Lane | Baltimore MD 21202 | 410.555.8175

James Smith

April 22, 2012

Page 2

Bearing Capacity

We used the most reliable procedure available, the XYZ method, to determine the soil's bearing capacity (i.e., its ability to withstand the weight of a loaded oil rig). That method required that we apply the following formula:

$Q = cN_v + tY$, where

Q = ultimate bearing capacity

c = average cohesive shear strength

N_v = the dimensionless bearing capacity factor

t = footing displacement

Y = weight of the soil unit

The final bearing capacity figure will be submitted in the final report, after we repeat the tests.

Seafloor Surface

By pulling our underwater seismometer back and forth across the project site, we developed a seismic "map" of the seafloor surface. That map seems typical of the flat floor expected in that area of the gulf. The only exception is the presence of what appears to be a small sunken boat. This wreck, however, is not in the immediate area of the proposed platform site.

CONCLUSIONS AND RECOMMENDATIONS

Based on our analysis, we conclude that there is only a slight risk of instability at the site. Although unlikely, it is possible that a rig leg could punch through the seafloor, either during or after loading. We base this opinion on (1) the existence of the weak clay layer, noted earlier, and (2) the marginal bearing capacity.

Nevertheless, we believe you can still place your platform if you follow careful rig-loading procedures. Specifically, take these precautions to reduce your risk:

1. Load the rig in 10-ton increments, waiting 1 hour between loadings.
2. Allow the rig to stand 24 hours after the loading and before placement of workers on board.
3. Have a soils specialist observe the entire loading process to assist with any emergency decisions if problems arise.

As noted at the outset, these conclusions and recommendations are based on preliminary data and analysis. We will complete our final study in three weeks and submit a formal report shortly thereafter.

M-Global, Inc., enjoyed working once again for Big Muddy Oil at its Gulf of Mexico lease holdings. I will phone you this week to see if you have any questions about our study. If you need information before then, please give me a call.

Sincerely,



Bartley Hopkins, Project Manager

M-Global, Inc.

hg

Notes why this method was chosen (i.e., reliability).

Explains both *how* the mapping procedure was done and *what* results it produced.

Leads off section with major conclusion, for emphasis.

Restates points (made in body) that support conclusion.

Uses list to emphasize recommendations to reduce risk.

Again mentions tentative nature of information, to prevent misuse of report.

Maintains contact and shows initiative by offering to call client.

TO: Gary Lane
 FROM: Jeff Bilstrom *JB*
 SUBJECT: Creation of Logo for Montrose Service Center
 DATE: October 3, 2012

Gives concise view of problem—and his proposed solution.

Part of my job as director of public relations is to get the Montrose name firmly entrenched in the minds of metro Atlanta residents. Having recently reviewed the contacts we have with the public, I believe we are sending a confusing message about the many services we offer retired citizens in this area.

To remedy the problem, I propose we adopt a logo to serve as an umbrella for all services and agencies supported by the Montrose Service Center. This proposal gives details about the problem and the proposed solution, including costs.

Includes effective lead-in.

The Problem

The lack of a logo presents a number of problems related to marketing the center's services and informing the public. Here are a few:

Uses bulleted list to highlight main difficulties posed by current situation.

- The letterhead mentions the organization's name in small type, with none of the impact that an accompanying logo would have.
- The current brochure needs the flair that could be provided by a logo on the cover page, rather than just the page of text and headings that we now have.
- Our 14 vehicles are difficult to identify because there is only the lettered organization name on the sides without any readily identifiable graphic.
- The sign in front of our campus, a main piece of free advertising, could better spread the word about Montrose if it contained a catchy logo.
- Other signs around campus could display the logo, as a way of reinforcing our identity and labeling buildings.

Ends section with good transition to next section.

It's clear that without a logo, the Montrose Service Center misses an excellent opportunity to educate the public about its services.

Starts with main point—need for logo.

The Solution

I believe a professionally designed logo could give the Montrose Service Center a more distinct identity. Helping to tie together all branches of our operation, it would give the public an easy-to-recognize symbol. As a result, there would be a stronger awareness of the center on the part of potential users and financial contributors.

Gary Lane
October 3, 2012
Page 2

The new logo could be used immediately to do the following:

- Design and print letterhead, envelopes, business cards, and a new brochure.
- Develop a decal for all company vehicles that would identify them as belonging to Montrose.
- Develop new signs for the entire campus, to include a new sign for the entrance to the campus, one sign at the entrance to the Blane Workshop, and one sign at the entrance to the Administration Building.

Focuses on *benefits* of proposed change.

Cost

Developing a new logo can be quite expensive. However, I have been able to get the name of a well-respected graphic artist in Atlanta who is willing to donate his services in the creation of a new logo. All that we must do is give him some general guidelines to follow and then choose among 8–10 rough sketches. Once a decision is made, the artist will provide a camera-ready copy of the new logo.

Emphasizes *benefit* of possible price break.

• Design charge	\$0.00
• Charge for new letterhead, envelopes, business cards, and brochures (min. order)	545.65
• Decal for vehicles 14 @ \$50.00 + 4%	728.00
• Signs for campus	415.28
Total Cost	<u>\$1,688.93</u>

Uses *listing* to clarify costs.

Conclusion

As the retirement population of Atlanta increases in the next few years, there will be a much greater need for the services of the Montrose Service Center. Because of that need, it's in our best interests to keep this growing market informed about the organization.

Closes with major benefit to reader and urge to action.

I'll stop by later this week to discuss any questions you might have about this proposal.

Keeps control of next step.

STUDY OF WILDWOOD CREEK

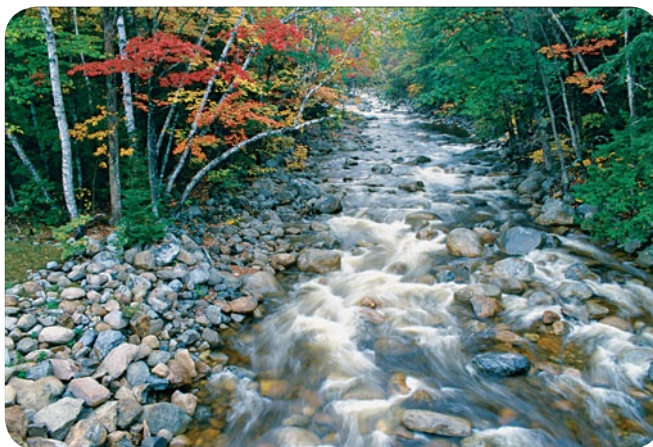
WINSLOW, GEORGIA

Prepared for:
The City of Winslow

Prepared by:
Christopher S. Rice, Hydro/Environmental Engineer
M-Global, Inc.

November 28, 2012

Uses graphic on title
page to reinforce theme
of environmental
protection. →



Medford Taylor/National Geographic Image Collection

12 Peachtree Street
Atlanta GA 30056
(404) 555-7524

McDuff Project #99-119
November 28, 2012

Adopt-a-Stream Program
City of Winslow
300 Lawrence Street
Winslow Georgia 30000

Attention: Ms. Elaine Sykes, Director

**STUDY OF WILDWOOD CREEK
WINSLOW, GEORGIA**

We have completed our seven-month project on the pollution study of Wildwood Creek. This project was authorized on May 16, 2012. We performed the study in accordance with our original proposal No. 14-P72, dated April 24, 2012.

This report mentions all completed tests and discusses the test results. Wildwood Creek scored well on many of the tests, but we are concerned about several problems—such as the level of phosphates in the stream. The few problems we observed during our study have led us to recommend that several additional tests should be completed.

Thank you for the opportunity to complete this project. We look forward to working with you on further tests for Wildwood Creek and other waterways in Winslow.

Sincerely,

Christopher S. Rice

Christopher S. Rice, P.E.
Hydro/Environmental Engineer

← Lists project title as it appears on title page.

← Gives brief statement of project information.

← Provides major point from report.



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Uses white space,
indenting, and bold to
accent organization
of report.

ILLUSTRATIONS**FIGURES**

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Includes illustration
titles as they appear
in text.

TABLES

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EXECUTIVE SUMMARY

Summarizes purpose
and scope of report.



The City of Winslow hired M-Global, Inc., to perform a pollution study of Wildwood Creek. The section of the creek that was studied is a one-mile-long area in Burns Nature Park, from Newell College to U.S. Highway 42. The study lasted seven months.

Describes major findings
and conclusions.



M-Global completed 13 tests on four different test dates. Wildwood scored fairly well on many of the tests, but there were some problem areas—for example, high levels of phosphates were uncovered in the water. The phosphates were derived either from fertilizer or from animal and plant matter and waste. Also uncovered were small numbers of undesirable water organisms that are tolerant of pollutants and can survive in harsh environments.

Includes main
recommendation
from report text.



M-Global recommends that (1) the tests done in this study be conducted two more times, through Spring 2013; (2) other environmental tests be conducted, as listed in the conclusions and recommendations section; and (3) a voluntary cleanup of the creek be scheduled. With these steps, we can better analyze the environmental integrity of Wildwood Creek.

INTRODUCTION

M-Global, Inc., has completed a follow-up to a study completed in 2004 by Ware County on the health of Wildwood Creek. This introduction describes the project site, scope of our study, and format for this report.

← Gives lead-in to Introduction.

PROJECT DESCRIPTION

By law, all states must clean up their waterways. The State of Georgia shares this responsibility with its counties. Ware County has certain waterways that are threatened and must be cleaned. Wildwood Creek is one of the more endangered waterways. The portion of the creek that was studied for this report is a one-mile stretch in the Burns Nature Park between Newell College and U.S. Highway 42.

← Briefly describes project.

SCOPE OF STUDY

The purpose of this project was to determine whether the health of the creek has changed since the previous study in 2004. Both physical and chemical tests were completed. The nine physical tests were as follows:

- Air temperature
- Water temperature
- Water flow
- Water appearance
- Habitat description
- Algae appearance
- Algae location
- Visible litter
- Bug count

← Uses bulleted list to emphasize scope of activities.

The four chemical tests were as follows:

- pH
- Dissolved oxygen (DO)
- Turbidity
- Phosphate

REPORT FORMAT

This report includes three main sections:

1. Field Investigation: A complete discussion of all the tests that were performed for the project
2. Test Comparison: Charts of the test results and comparisons
3. Conclusions and Recommendations

← Provides “map” of main sections in report.

FIELD INVESTIGATION

Wildwood Creek has been cited repeatedly for environmental violations in the pollution of its water. Many factors can generate pollution and affect the overall health of the creek. In 2004, the creek was studied in the context of a study of all water systems in Ware County. Wildwood Creek was determined to be one of the more threatened creeks in the county.

The city needed to learn if much has changed in the past eight years, so M-Global was hired to perform a variety of tests on the creek. Our effort involved a more in-depth study than that done in 2004. Tests were conducted four times over a seven-month period. The 2004 study lasted only one day.

The field investigation included two categories of tests: physical tests and chemical tests.

PHYSICAL TESTS

The physical tests covered a broad range of environmental features. This section discusses the importance of the tests and some major findings. The Test Comparison section on Page 9 includes a table that lists results of the tests and the completion dates. The test types were as follows: air temperature, water temperature, water flow, water appearance, habitat description, algae appearance, algae location, visible litter, and bug count.

Air Temperature

The temperature of the air surrounding the creek will affect life in the water. Unusual air temperature for the seasons will determine if life can grow in or out of the water.

Three of the four tests were performed in the warmer months. Only one was completed on a cool day. The difference in temperature from the warmest to coolest day was 10.5°C, an acceptable range.

Water Temperature

The temperature of the water determines which species will be present. Also affected are the feeding, reproduction, and metabolism of these species. If there are one or two weeks of high temperature, the stream is unsuitable for most species. If water temperature changes more than 1° to 2°C in 24 hours, thermal stress and shock can occur, killing much of the life in the creek.

During our study, the temperature of the water averaged 1°C cooler than the temperature of the air. The water temperature did not get above 23°C or below 13°C. These ranges are acceptable by law.

Amplifies information
presented later in report.

Water Flow

The flow of the water influences the type of life in the stream. Periods of high flow can cause erosion to occur on the banks and sediment to cover the streambed. Low water flow can decrease the living space and deplete the oxygen supply.

The flow of water was at the correct level for the times of year the tests were done—except for June, which had a high rainfall. With continual rain and sudden flash floods, the creek was almost too dangerous for the study to be performed that month.

In fact, in June we witnessed the aftermath of one flash flood. Figure 1 shows the creek with an average flow of water, and Figure 2 shows the creek during the flood. The water's average depth is 10 inches. During the flash flood, the water level rose and fell 10 feet in about one hour. Much dirt and debris were washing into the creek, and some small fish were left on dry land as the water receded.

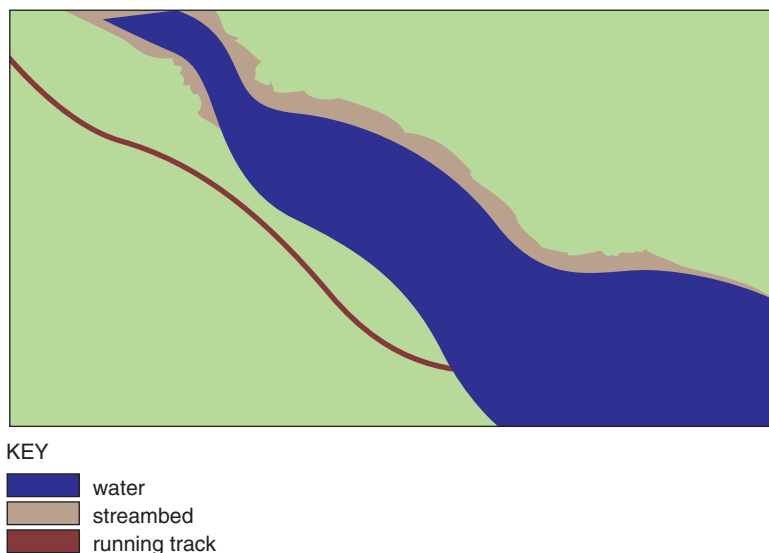


KEY

- water
- streambed
- running track

■ **Figure 1** ■ Wildwood Creek—Normal Water Level

← Incorporates graphic into page of text.



■ **Figure 2** ■ Wildwood Creek—Flash Flood Water Level

Water Appearance

The color of the water gives a quick but fairly accurate view of the health of the creek. If the water is brown or dirty, then silt or human waste may be present. Black areas of water may contain oil or other chemical products.

On each of the four test days, the water was always clear. Therefore the appearance of the creek water was considered excellent.

Habitat Description

The habitat description concerns the appearance of the stream and its surroundings. An important criterion is the number of pools and the number of ripples—that is, points where water flows quickly over a rocky area. Both pools and ripples provide good locations for fish and other stream creatures to live and breed.

In describing habitat, M-Global also evaluates the amount of sediment at the bottom of the stream. Too much sediment tends to cover up areas where aquatic life lays eggs and hides them from predators. We also evaluate the stability of the stream banks; a stable bank indicates that erosion has not damaged the habitat. Finally, we observe the amount of stream cover. Such vegetation helps keep soil in place on the banks.

7

Wildwood Creek tested fairly well for habitat. The number of pools and ripples was about average for such creeks. Stream deposits and stream bank stability were average to good, and stream cover was good to excellent. For more detail about test results, see the chart in the Test Comparison section on Page 9.

Algae Appearance and Location

Algae are naturally present in any creek. The numbers of algae can be a warning of pollution in the water. If algae are growing out of control, disproportionate amounts of nutrients such as nitrogen or phosphate could be present. These chemicals could come from fertilizer washed into the creek. Excessive numbers of algae cause the oxygen level to drop when they die and decompose.

During the four studies, algae were everywhere, but they were especially heavy on the rocks in the ripples of the creek. The algae were always brown and sometimes hairy.

Visible Litter

Litter can affect the habitat of a creek. Although some litter has chemicals that can pollute the water, other litter can cover nesting areas and suffocate small animals. Whether the litter is harmful or not, it is always an eyesore.

On all four test dates, the litter we saw was heavy and ranged from tires to plastic bags. Some of the same trash that was at the site on the first visit was still there seven months later.

Gives specific details that support the report's conclusions and recommendations, which come later.

Bug Count

The bug count is a procedure that begins by washing dirt and water onto a screen. As water drains, the dirt with organisms is left on the screen. The bugs are removed and classified. Generally, the lower the bug count, the higher the pollution levels. Bug counts here were considered low to average.

Two types of aquatic worms were discovered every time during our count, but in relatively small amounts. In addition, the worms we observed are very tolerant of pollution and can live in most conditions. Finally, we observed only two crayfish, animals that are somewhat sensitive to pollution.

CHEMICAL TESTS

Physical tests cover areas seen with the naked eye. Chemical tests can uncover pollutants that are not so recognizable. Certain chemicals can wipe out all life in a creek. Other chemicals can cause an overabundance of one life-form, which in turn could kill more sensitive animals.

A chart of results of chemical tests is included in the Test Comparison section on Page 9. The chemical tests that M-Global performed were pH, dissolved oxygen (DO), turbidity, and phosphate.

pH

The pH test is a measure of active hydrogen ions in a sample. The range of the pH test is 0–14. If the sample is in the range of 0–7.0, it is acidic; if the sample is in the range of 7.0–14, it is basic. By law, the pH of a water sample must be within the range of 6.0–8.5.

For the tests we completed, the water sample was always 7.0, which is very good for a creek.

Dissolved Oxygen (DO)

Normally, oxygen dissolves readily into water from surface air. Once dissolved, it diffuses slowly in the water and is distributed throughout the creek. The amount of DO depends on the circumstances. Oxygen is always highest in choppy water, just after noon, and in cooler temperatures.

In many streams, the level of DO can become critically low during the summer months. When the temperature is warm, organisms are highly active and consume the oxygen supply. If the amount of DO drops below 3.0 ppm (parts per million), the area can become stressful for the organisms. An amount of oxygen that is 2.0 ppm or below will not support fish. DO that is 5.0 ppm to 6.0 ppm is usually required for growth and activity of organisms in the water.

According to the Water Quality Criteria for Georgia, average daily amounts of DO should be 5.0 ppm, with a minimum of 4.0 ppm. Wildwood Creek scored well on this test. The average amount of DO in the water was 6.9 ppm, with the highest amount being 9.0 ppm on November 19, 2012.

Turbidity

Turbidity is the discoloration of water due to sediment, microscopic organisms, and other matter. One major factor of turbidity is the level of rainfall before a test.

Three of our tests were performed on clear days with little rainfall. On these dates, the turbidity of Wildwood Creek was always 1.0, the best that creek water can score on the test. The fourth test, which scored worse, occurred during a rainy period.

Phosphate

Phosphorus occurs naturally as phosphates—for example, orthophosphates and organically bound phosphates. Orthophosphates are phosphates that are formed in fertilizer, whereas organically bound phosphates can form in plant and animal matter and waste.

Phosphate levels higher than 0.03 ppm contribute to an increase in plant growth. If phosphate levels are above 0.1 ppm, plants may be stimulated to grow out of control. The phosphate level of Wildwood was always 0.5 ppm, considerably higher than is desirable.

TEST COMPARISON

There was little change from each of the four test dates. The only tests that varied greatly from one test to another were air temperature, water temperature, water flow, and DO. On the basis of these results, it would appear that Wildwood Creek is a relatively stable environment.

Table 1 Physical Tests

Brings together test results for easy reference.

TEST DATES	5/26/12	6/25/12	9/24/12	11/19/12
Air Temperature in °C	21.5	23.0	24.0	13.5
Water Temperature in °C	20.0	22.0	23.0	13.0
Water Flow	Normal	High	Normal	Normal
Water Appearance	Clear	Clear	Clear	Clear
Habitat Description				
Number of Pools	2.0	3.0	2.0	5.0
Number of Ripples	1.0	2.0	2.0	2.0
Amount of Sediment Deposit	Average	Average	Good	Average
Stream Bank Stability	Average	Good	Good	Good
Stream Cover	Excellent	Good	Excellent	Good
Algae Appearance	Brown	Brown/hairy	Brown	Brown
Algae Location	Everywhere	Everywhere	Attached	Everywhere
Visible Litter	Heavy	Heavy	Heavy	Heavy
Bug Count	Low	Average	Low	Average

Table 2 Chemical Tests

Test	5/26/12	6/25/12	9/24/12	11/19/12
pH	7.0	7.0	7.0	7.0
Dissolved Oxygen (DO)	6.8	6.0	5.6	9.0
Turbidity	1.0	3.0	1.0	1.0
Phosphate	0.50	0.50	0.50	0.50

CONCLUSIONS AND RECOMMENDATIONS

This section includes the major conclusions and recommendations from our study of Wildwood Creek.

CONCLUSIONS

Generally, we were pleased with the health of the stream bank and its floodplain. The area studied has large amounts of vegetation along the stream, and the banks seem to be sturdy. The floodplain has been turned into a park, which handles floods in a natural way. Floodwater in this area comes in contact with vegetation and some dirt. Floodwater also drains quickly, which keeps sediment from building up in the creek.

However, we are concerned about the number and types of animals uncovered in our bug counts. Only two bug types were discovered, and these were types quite tolerant of pollutants. The time of year these tests were performed could have affected the discovery of some animals. However, the low count still should be considered a possible warning sign about water quality. Phosphate levels were also high and probably are the cause of the large numbers of algae.

We believe something in the water is keeping sensitive animals from developing. One factor that affects the number of animals discovered is the pollutant problems in the past (see Appendix A). The creek may still be in a redevelopment stage, a possible explanation for the small numbers of animals.

RECOMMENDATIONS

On the basis of these conclusions, we recommend the following actions for Wildwood Creek:

1. Conduct the current tests two more times, through Spring 2013. Spring is the time of year that most aquatic insects are hatched. If sensitive organisms are found then, the health of the creek could be considered to have improved.
2. Add testing for nitrogen. With the phosphate level being so high, nitrogen might also be present. If it is, then fertilizer could be in the water.
3. Add testing for human waste. Some contamination may still be occurring.
4. Add testing for metals, such as mercury, that can pollute the water.
5. Add testing for runoff water from drainage pipes that flow into the creek.
6. Schedule a volunteer cleanup of the creek.

With a full year of study and additional tests, the problems of Wildwood Creek can be better understood.

Draws conclusions that flow from data in body of report.

Uses paragraph format instead of lists because of lengthy explanations needed.

Gives numbered list of recommendations for easy reference.

APPENDIX A

Background on Wildwood Creek

Wildwood Creek begins from tributaries on the northeast side of the city of Winslow. From this point, the creek flows southwest to the Chattahoochee River. Winslow Wastewater Treatment Plant has severely polluted the creek in the past with discharge of wastewater directly into the creek. Wildwood became so contaminated that signs warning of excessive pollution were posted along the creek to alert the public.

Today, all known wastewater discharge has been removed. The stream's condition has dramatically improved, but nonpoint contamination sources continue to lower the creek's water quality. Nonpoint contamination includes sewer breaks, chemical dumping, and storm sewers.

Another problem for Wildwood Creek is siltration. Rainfall combines with bank erosion and habitat destruction to wash excess dirt into the creek. This harsh action destroys most of the macroinvertebrates. At the present time, Wildwood Creek may be one of the more threatened creeks in Ware County.

APPENDIX B

Water Quality Criteria for Georgia

All waterways in Georgia are classified in one of the following categories: fishing, recreation, drinking, and wild and scenic. Different protection levels apply to the different uses. For example, the protection level for dissolved oxygen is stricter in drinking water than fishing water. All water is supposed to be free from all types of waste and sewage that can settle and form sludge deposits.

In Ware County, all waterways are classified as “fishing,” according to Chapter 391-3-6.03 of “Water Use Classifications and Water Quality Standards” in the Georgia Department of Natural Resources *Rules and Regulations for Water Quality Control*. The only exception is the Chattahoochee River, which is classified as “drinking water supply” and “recreational.”

APPENDIX C

Map 6
Location of City of Winslow
Parks and Recreation Facilities

